

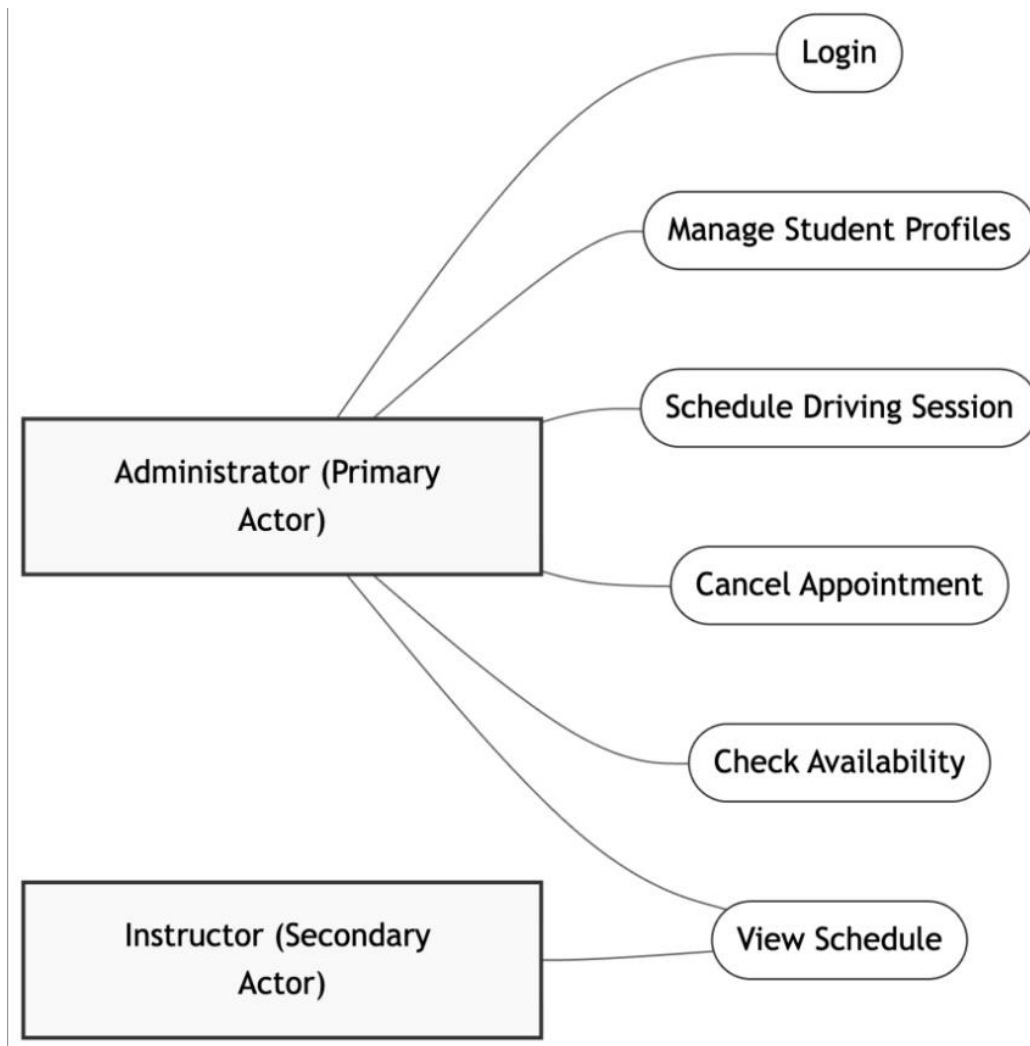
Nicole Snyder

02-20-2026

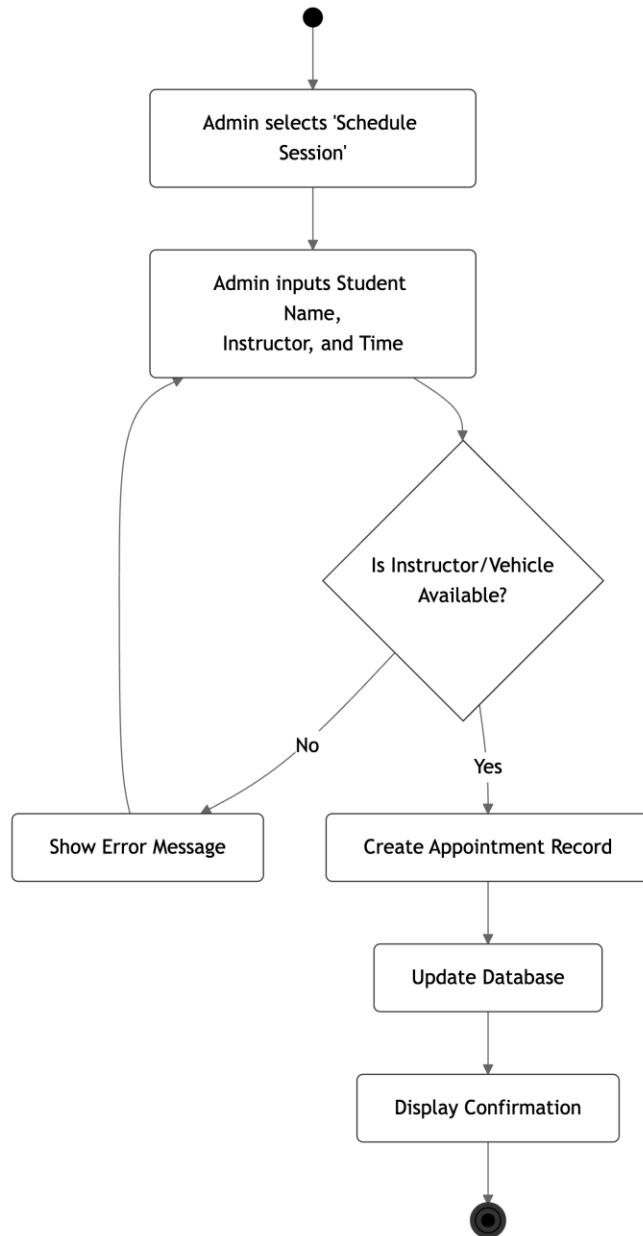
Professor Matthew McHann

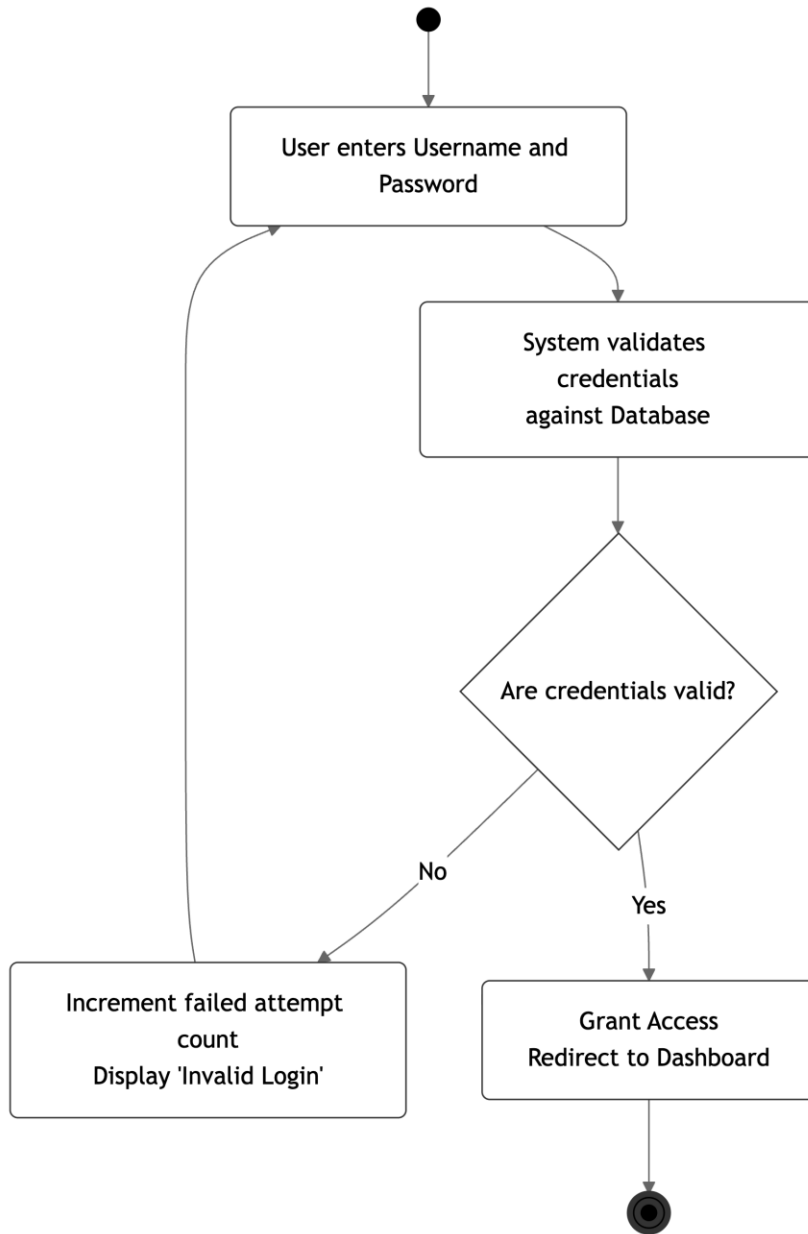
UML Diagrams

UML Use Case Diagram

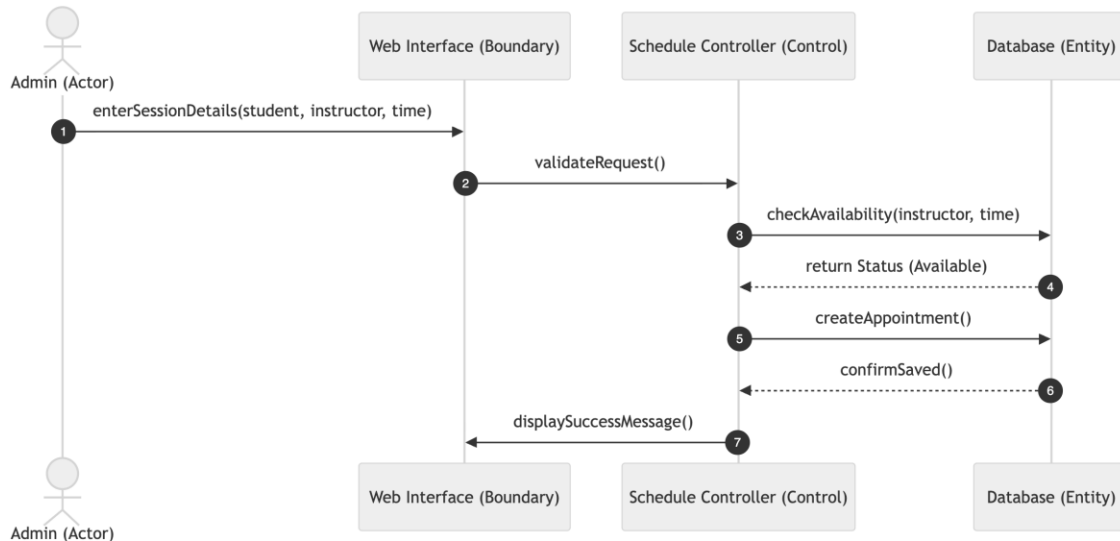


UML Activity Diagrams

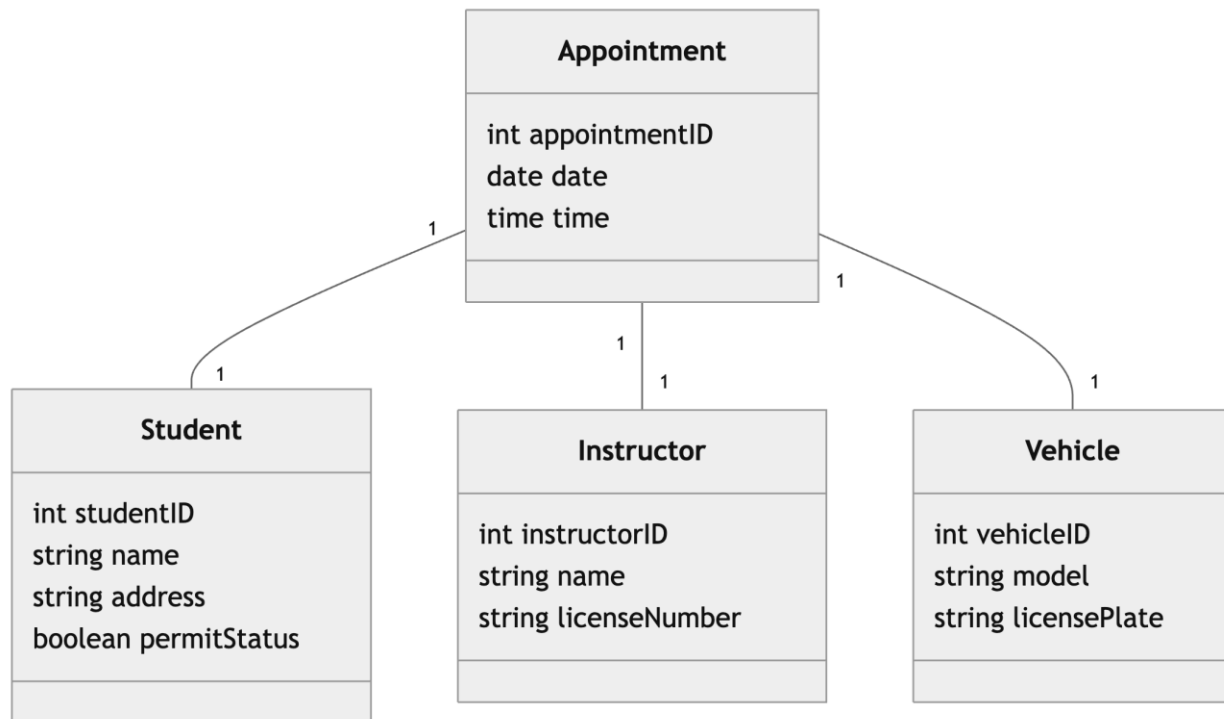




UML Sequence Diagram



UML Class Diagram



Technical Requirements

Based on the functional and nonfunctional requirements outlined in Project One, the DriverPass system requires the following technical infrastructure:

- Hardware Requirements:**
 - Client Side:** Standard office desktop or laptop computers capable of running modern web browsers (e.g., 8GB RAM, i5 Processor). No specialized hardware is required for the users.
 - Server Side:** A web server capable of handling concurrent requests. The server must host the Java runtime environment and the database backend.

- **Software Requirements:**
 - **Operating System:** Cross-platform compatibility for clients (Windows, macOS) via web browser. The server will likely run on a Linux-based environment (e.g., Ubuntu) for stability.
 - **Database:** A relational database management system (RDBMS) such as MySQL or PostgreSQL is required to manage the relationships between students, instructors, and schedules.
 - **Development Framework:** The backend will be developed using Java (consistent with the class structure), utilizing a web framework like Spring Boot. The frontend will utilize HTML5 and CSS for the user interface.
- **Tools & Infrastructure:**
 - **Network:** A reliable high-speed internet connection is required for real-time database updates.
 - **Security:** An SSL/HTTPS certificate is required to encrypt data transmission between the office computers and the web server to protect student privacy.